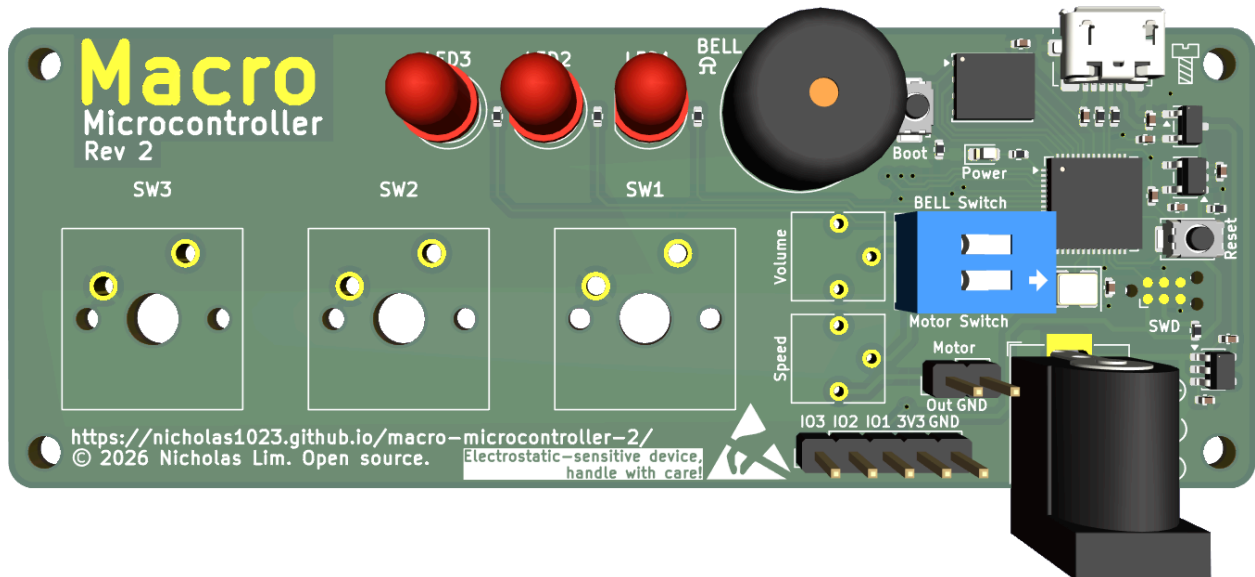


# Macro Microcontroller 2 Datasheet



## Overview

Macro Microcontroller 2 is an RP2040 microcontroller development board with an integrated macropad that lets you control GPIOs with keyboard keys on a single PCB, plus a custom BASIC interpreter and disk operating system.

## Specifications

Microcontroller Unit: RP2040

Power: 5V input via data port or barrel jack, 3.3V input via GPIO pin headers, maximum logic voltage is 3.3V.

Dimensions: 96.52 × 35.56 mm

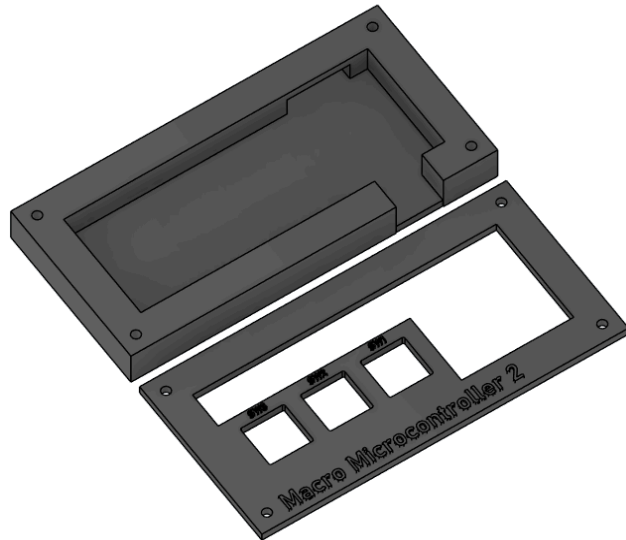
Flash Memory: 16 MB

Mounting Hole Dimensions: 2.5mm diameter

## Warnings

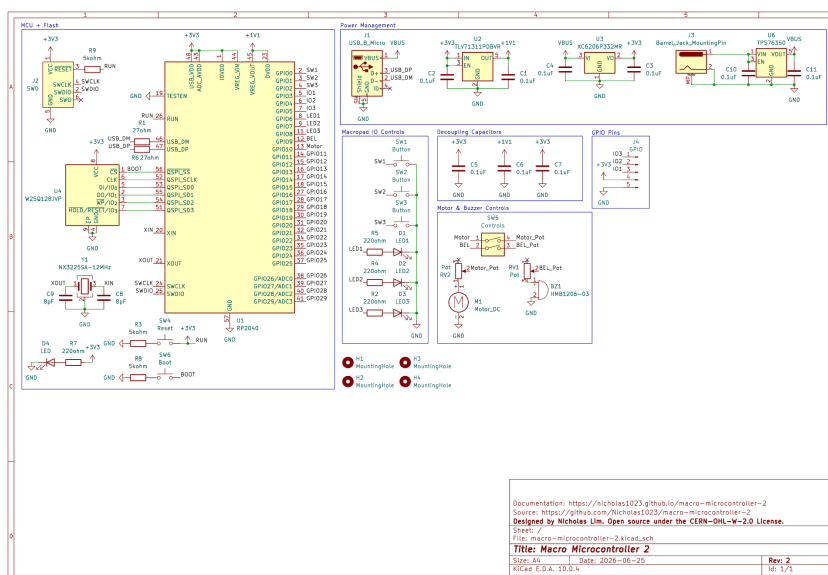
Components may be damaged if the logic voltage applied is more than 3.3V or more than one power source is used at the same time. Use only one power source, either via the data port, barrel jack or GPIO pin header to power the board. It is safe to use 5V via the data port and barrel jack as low dropout regulators are connected to reduce the

# Enclosure

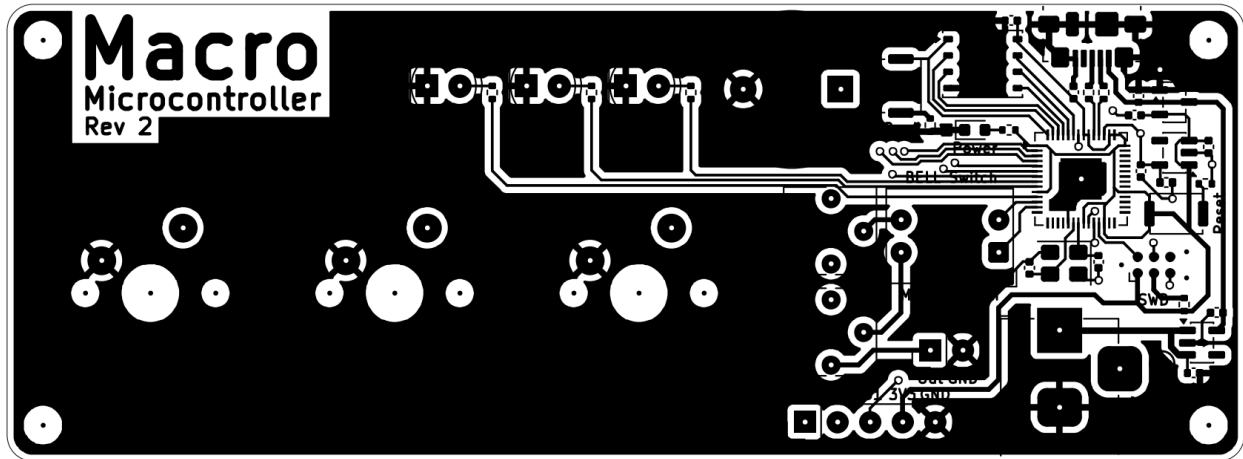


<https://github.com/Nicholas1023/macro-microcontroller-2/releases/download/v0.0.1/macro-microcontroller-2-enclosure.stl>

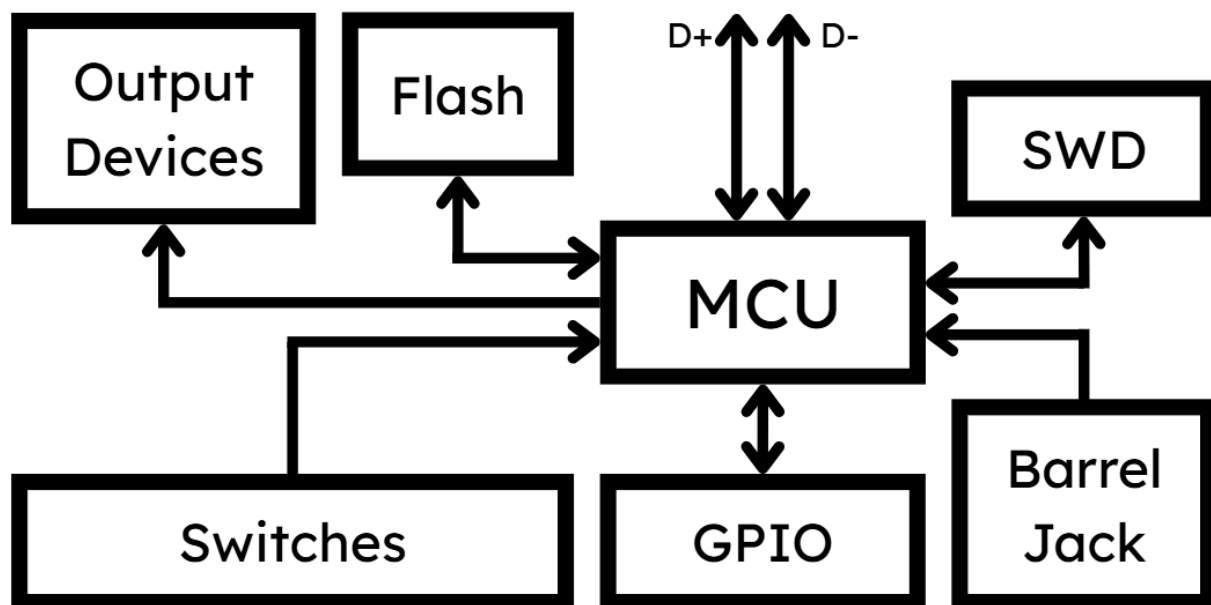
## Schematics



## PCB Routing and Layout



## Block Diagram



## Pinout

Macro Microcontroller 2 provides a few IO pinouts for connecting peripherals.

- DC Motor (MOTOR)
- 2.54mm GPIO pin headers (IO3, IO2, IO1, 3.3V, GND)

| Macro Microcontroller 2 Pin # | Corresponding RP2040 Pin # |
|-------------------------------|----------------------------|
| IO1                           | GPIO3                      |
| IO2                           | GPIO4                      |
| IO3                           | GPIO5                      |

## Built-in IO Devices

- Key switches (SW1, SW2, SW3)
- LEDs (LED1, LED2, LED3)
- Buzzer (BELL)

## Bill of Materials

| Qty | Reference Designator                 | Manufacturer Part # |
|-----|--------------------------------------|---------------------|
| 1   | BZ1                                  | HMB1206-03          |
| 1   | J3                                   | DC-005 2.0          |
| 1   | SW5                                  | DS-02RP             |
| 1   | M1                                   | 2.54 1x2 Pin Header |
| 1   | J4                                   | 2.54 1x5 Pin Header |
| 1   | J1                                   | 473460001           |
| 9   | C1, C10, C11, C2, C3, C4, C5, C6, C7 | 0.1µF 0402          |
| 1   | U1                                   | RP2040              |
| 1   | U3                                   | XC6206P332MR        |
| 1   | U2                                   | TLV71311PDBVR       |
| 4   | R2, R4, R5, R7                       | 220Ω 0402           |
| 1   | D4                                   | NCD0603B1           |
| 2   | C8, C9                               | 8pF 0402            |
| 1   | U4                                   | W25Q128JVPIQ        |

|   |               |                          |
|---|---------------|--------------------------|
| 3 | D1, D2, D3    | 333-2SURD/S530-A3        |
| 1 | U6            | TPS76350DBVR             |
| 1 | Y1            | NX3225SA-12MHz-STD-CSR-6 |
| 2 | R1, R6        | 27Ω 0402                 |
| 3 | R3, R8, R9    | 5kΩ 0402                 |
| 2 | RV1, RV2      | T73YP202KT20             |
| 3 | SW1, SW2, SW3 | MX1A-E1NA                |
| 2 | SW4, SW6      | SKRKAEE020               |

## Macro Microcontroller DOS

Macro Microcontroller Disk Operating System (DOS) is a simple command-line operating system providing basic programs for Macro Microcontroller 2.

### Installation

1. Download [macro-microcontroller-dos-0.0.2.uf2](#). This release includes Macro Microcontroller BIOS, A basic BIOS-like firmware, which performs POST routines, reports information on the device and boot the DOS.

SHA-256 Checksum:

a3c7d595ed9463521bcf292c6aa4192db7e512f84110ece2724c2aba2e7f2fc3

2. Plug in Macro Microcontroller 2 while holding the Boot button. A storage drive will appear on your computer.
3. Move macro-microcontroller-dos-0.0.2.uf2 into the storage drive. Your development board is now running Macro Microcontroller BIOS and DOS.
4. Install a serial terminal application on your computer and connect it to the board. If successful, you should see the following:

```
|_/_/_| Macro Microcontroller BIOS Version 0.0.2.  
|_/_/_| Copyright (C) 2025-2026 Nicholas Lim.
```

```
Device: Macro Microcontroller 2 (RP2040 @ 125MHz)  
Serial Number: *****  
RAM Free: 254K OK  
Serial Status: Connected
```

```
Press SW1 to enter Macro Microcontroller BASIC.  
Press SW2 to enter BOOTSEL mode.
```

```
Booting system...
```

```
Macro Microcontroller Disk Operating System Version 0.0.2.  
Copyright (C) 2026 Nicholas Lim.  
Drive A>
```

## Commands and Programs

- **HELP:** Launches the help guide.
- **CALCULATE:** Performs basic operations on 2 numbers. Usage: `CALCULATE [Number] [+ - * / ^ root] [Number]`
- **ECHO:** Echoes text.
- **LS:** Show program listing.
- **SYSTEM:** Starts another Macro Microcontroller DOS console.
- **EXIT:** Stops all processes for power off.
- **GPIO:** Turns on a specified GPIO for 1 second. Usage: `GPIO [LED# IO# BELL MOTOR]`
- **VERSION:** Shows the release version.
- **REBOOT:** Reboots the system. Usage: `REBOOT -m [BASIC BOOTSEL]*`
- **INFO:** Displays device and memory information.
- **UPTIME:** Shows device uptime.

## Predefined Values for Parameters

- **IO\*:** Set GPIO pins 1, 2 or 3 to high.
- **LED\*:** Set LEDs 1, 2 or 3 to high.
- **BELL:** Sounds the builtin buzzer.
- **MOTOR:** Sets motor pins to high.
- **BASIC:** Reboot in BASIC interpreter mode.
- **BOOTSEL:** Reboot in BOOTSEL mode.

# Macro Microcontroller BASIC

Macro Microcontroller BASIC is a custom BASIC interpreter written in C with low-level hardware communication, providing a programming environment via a serial terminal.

## Installation

1. Download [macro-microcontroller-basic-0.0.3.uf2](#).  
SHA-256 Checksum:  
96d4aa58d61d796067c1668cc37896fa968a55dceea188e50f26e067d116e99f
2. Plug in Macro Microcontroller 2 while holding the Boot button. A storage drive will appear on your computer.
3. Move macro-microcontroller-basic-0.0.3.uf2 into the storage drive. Your development board is now running Macro Microcontroller BASIC.
4. Install a serial terminal application on your computer and connect it to the board. If successful, you should see the following:  

```
Macro Microcontroller BASIC Version 0.0.3.  
Copyright (C) 2025-2026 Nicholas Lim.  
***** bytes of RAM free.  
Ready.  
10
```
5. If not, try sending `PRINT "It works!"`. If "It works!" is sent back to the serial terminal, you have successfully installed Macro Microcontroller BASIC.

## Statements

- **PRINT:** Outputs text to the serial terminal.
- **LET:** Assigns a value (integer, float or string) to a variable.
- **INPUT:** Assigns a value (integer, float or string) to a variable via user input.
- **GPIO:** Turns on a specified GPIO pin for 1 second.
- **LSVAR:** Outputs variable name and their value to the serial terminal.
- **END:** Clears variable and reset line numbers.
- **EXIT:** Puts the device into sleep/power off mode.
- **KEY:** Pass GPIO controls to the macropad.
- **REM:** Add comments to your program.

## Predefined Values

- **IO\*:** Set GPIO pins 1, 2 or 3 to high.
- **LED\*:** Set LEDs 1, 2 or 3 to high.
- **BELL:** Sounds the builtin buzzer.
- **MOTOR:** Sets motor pins to high.
- **RND:** Generates a random floating-point integer less than 1.

For more information on Macro Microcontroller 2, visit:

<https://github.com/Nicholas1023/macro-microcontroller-2> (Repository)

<https://nicholas1023.github.io/macro-microcontroller-2> (Documentation site)

## Licensing

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